

OPTIMIZATION AND ITS APPLICATIONS

OPTIMIZATION THEORY AND METHODS

Nonlinear Programming

WENYU SUN
YA-XIANG YUAN

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Nonlinear Programming

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Aims and Scope

Optimization has been expanding in all directions at an astonishing rate during the last few decades. New algorithmic and theoretical techniques have been developed, the diffusion into other disciplines has proceeded at a rapid pace, and our knowledge of all aspects of the field has grown even more profound. At the same time, one of the most striking trends in optimization is the constantly increasing emphasis on the interdisciplinary nature of the field. Optimization has been a basic tool in all areas of applied mathematics, engineering, medicine, economics and other sciences.

The series *Springer Optimization and Its Applications* publishes undergraduate and graduate textbooks, monographs and state-of-the-art expository works that focus on algorithms for solving optimization problems and also study applications involving such problems. Some of the topics covered include nonlinear optimization (convex and nonconvex), network flow problems, stochastic optimization, optimal control, discrete optimization, multi-objective programming, description of software packages, approximation techniques and heuristic approaches.

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By

WENYU SUN

Nanjing Normal University, Nanjing, China

YA-XIANG YUAN

Chinese Academy of Science, Beijing, China



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Preface

Optimization is a subject that is widely and increasingly used in science, engineering, economics, management, industry, and other areas. It deals with selecting the best of many possible decisions in real-life environment, constructing computational methods to find optimal solutions, exploring the theoretical properties, and studying the computational performance of numerical algorithms implemented based on computational methods.

Along with the rapid development of high-performance computers and progress of computational methods, more and more large-scale optimization problems have been studied and solved. As pointed out by Professor Yuqi He of Harvard University, a member of the US National Academy of Engineering, optimization is a cornerstone for the development of civilization.

This book systematically introduces optimization theory and methods, discusses in detail optimality conditions, and develops computational methods for unconstrained, constrained, and nonsmooth optimization. Due to limited space, we do not cover all important topics in optimization. We omit some important topics, such as linear programming, conic convex programming, mathematical programming with equilibrium constraints, semi-infinite programming, and global optimization. Interested readers can refer to Dantzig [78], Walsch [347], Shu-Cheng Fang and S. Puthenpura [121], Luo, Pang, and Ralph [202], Wright [358], Wolkowitz, Saigal, and Vandenbergh [355].

The book contains a lot of recent research results on nonlinear programming including those of the authors, for example, results on trust region methods, inexact Newton method, self-scaling variable metric method, conic model method, non-quasi-Newton method, sequential quadratic programming, and nonsmooth optimization, etc.. We have tried to make the book

self-contained, systematic in theory and algorithms, and easy to read. For most methods, we motivate the idea, study the derivation, establish the global and local convergence, and indicate the efficiency and reliability of the numerical performance. The book also contains an extensive, not complete, bibliography which is an important part of the book, and the authors hope that it will be useful to readers for their further studies.

This book is a result of our teaching experience in various universities and institutes in China and Brazil in the past ten years. It can be used as a textbook for an optimization course for graduates and senior undergraduates in mathematics, computational and applied mathematics, computer science, operations research, science and engineering. It can also be used as a reference book for researchers and engineers.

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Wenyu Sun, Nanjing Normal University
Yaxiang Yuan, Chinese Academy of Science
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